which is, as we know, a division algebra. The proof of this generalized version of Proposition 3.1.4 is the same as before (check it!).

Here is an alternative proof of Proposition 3.1.4.[2]

By Remark 3.1.3, if $V = \oplus_X V_X \otimes X$ and $U = \oplus_X U_X \otimes X$ for some vector spaces V_X and U_X, then we have a natural isomorphism

$$\mathrm{Hom}_A(V, U) \cong \prod_X \mathrm{Hom}(V_X, U_X).$$

Now let $f\colon V \to U$ correspond to the tuple $(f_X\colon V_X \to U_X)$. Then f is injective (respectively surjective, an isomorphism) if and only if all the f_X are.

Now, suppose $V = \oplus_{i\in I} V_i$ with V_i irreducible, and $f : V \to U$ is a surjective homomorphism.

Lemma 3.1.6. *There exists a subset $J \subseteq I$ such that $V_J := \oplus_{i\in J} V_i$ is mapped isomorphically by f onto U.*

Proof. Let J be a maximal subset such that $f|_{V_J}$ is injective. If $f(V_J) \neq U$, then there exists $i \in I$ such that $f(V_i)$ is not contained in $f(V_J)$. Then the map $V_i \to U/f(V_J)$ is nonzero, and hence injective by Schur's lemma. Let $J' = J \cup \{i\}$; then f is injective on $V_{J'}$, contradicting the maximality of J, which proves the lemma. □

Now we are ready to prove Proposition 3.1.4. Let W be a submodule of $V := \oplus_X V_X \otimes X$, where $\dim V < \infty$. We claim that $W = \oplus_X W_X \otimes X$ for some vector spaces $W_X \subseteq V_X$. Indeed, by Lemma 3.1.6 have $V/W = \oplus_X U_X \otimes X$ for some vector spaces U_X, so

$$W = \mathrm{Ker}(V \to V/W) = \oplus_X \mathrm{Ker}(V_X \to U_X) \otimes X,$$

as desired.

3.2. The density theorem

Let A be an algebra over an algebraically closed field k.

[2]We thank B. Poonen for this argument